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The Cabinet Committee on Security has flagged off the prototype development and testing stage of India's indigenous fifth generation stealth fighter jet, the Advanced Medium Combat Aircraft (AMCA). Slated to be in production from 2035, it would make the IAF battle-ready for the future

By Pradip R. Sagar



SOUNDLESS STRIKE A model of the AMCA displayed at the DefExpo 2022 in Gandhinagar

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With air power being regarded as the sword arm of offensive operations, as well as crucial in defence, in all modern military doctrines, developing more sophisticated fighter jets has been a continuing objective for top military powers. The term 'fifth generation fighter aircraft' (FGFA) is currently used to denote the few most technologically advanced fighter jets—a super elite category—that have been developed over the past 30 years. Such machines have multirole capability, advanced avionics and networked data fusion (enabling greater situation awareness on the battlefield). But the one quality that sets a fifth generation jet apart from other modern fighters is its 'stealth' capability—the quality of being unseen and undetected by enemy's ground defence radars and aircraft. As can be imagined, making a FGFA is a complex task, and only a few fighters inhabit that class—the Lockheed Martin F-35 Lightning II and F22 Raptor (US), the Chengdu J-20 Mighty Dragon (China) and the Sukhoi Su-57 Felon (Russia). On February 21, Turkey conducted the maiden test flight of its indigenously developed fifth generation stealth fighter jet, Kaan. It is developed by Turkish Aerospace Industries and BAE Systems of the UK.

Here in India, aerospace engineers at the Aeronautical Development Agency (ADA) complex in Vimanpura, an aeronautical hub in Bengaluru, have been quietly working on designing India's own fifth generation stealth jet—the Advanced Medium Combat Aircraft (AMCA)—since 2009. Now, in a push that could propel India into the exclusive club of nations that operate FGFAs, the Cabinet Committee on Security (CCS), the country's highest body on security matters, gave its nod to the next stage of the AMCA project on March 7 by sanctioning Rs 15,000 crore for its prototype development. The five twin-engine prototypes would be made by the ADA—which functions under the Defence Research and Development Organisation (DRDO) of the Ministry of Defence (MoD)—and Hin-